TRAINING REPORT

On

Secure Employee Data Visualization System

Defence Research and Development Laboratory (DRDL)
Kanchanbagh, Hyderabad – 500058

In partial fulfilment of the degree

MASTER OF TECHNOLOGY

In

Computer Science & Engineering (CNIS)

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Visakhapatnam, Andhra Pradesh-530003

Submitted By

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Under the guidance of

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DECLARATION

I, **Bayana Aishwarya**, hereby declare that this report is being submitted in partial fulfilment of the requirements of my internship at **Defence Research and Development Laboratory (DRDL).**

This report is an original work carried out by me during my internship and has been completed under the guidance of **Shri B V. Hari Krishna Nanda (Scientist 'G')** and **Smt. G M V. Lakshmi**. This internship was undertaken as a part of the academic curriculum of Andhra University College of Engineering, Visakhapatnam, Andhra Pradesh-530003.

Signature

Smt. G M V Lakshmi

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Shri B V. Hari Krishna Nanda

(Scientist 'G')

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I also wish to express my gratitude to all the faculty members of the Department of Computer Science and Engineering (CNIS) of **Andhra University College of Engineering** for their encouragement and support during this training.

Finally, I would like to thank the library and all other staff members who provided access to essential resources that greatly aided my learning during the course of this training.

Signature

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Defence Research and Development Laboratory (DRDL), located in Hyderabad, is a premier laboratory engaged in the research and development of missile systems for the Indian Armed Forces. DRDL focuses on the design, development, and integration of advanced missile technologies to enhance national defence capabilities.

The laboratory works in key areas such as missile propulsion, guidance and control, avionics, structures, and system integration. DRDL is equipped with modern facilities and advanced infrastructure to support high-quality research, testing, and development activities.

DRDL also provides opportunities for students to gain practical exposure through internship programs, enabling them to understand professional research practices, teamwork, and discipline in a defence organisation.

HISTORY

The Defence Research and Development Laboratory (DRDL) was established in 1961 under the Defence Research and Development Organisation (DRDO) with the primary objective of developing indigenous missile systems for India's defence forces. Located at Kanchanbagh, Hyderabad, DRDL marked the beginning of India's journey towards self-reliance in missile technology during an era when the country was largely dependent on foreign defence equipment.

In its formative years, DRDL initiated several pivotal programmes such as Project Indigo, a surface-to-air missile project which, although not fully successful, laid the groundwork for more advanced initiatives. This was followed by Project Devil and Project Valiant in the 1970s, which contributed significantly to the development of short-range missile technologies and laid a strong foundation for future projects.

The laboratory's efforts culminated in the launch of the Integrated Guided Missile Development Programme (IGMDP) in the 1980s, a major milestone in India's missile development history. DRDL played a critical role in the successful design and development of key strategic missile systems under this programme, including Prithvi, Agni, Akash, and Nag.

Over the years, DRDL has evolved into a premier centre of excellence in missile technology, specialising in areas such as aerodynamics, propulsion, guidance and control, warhead technology, and missile systems integration. With state-of-the-art infrastructure for design, simulation, testing, and validation, DRDL supports end-to-end development of both technical and strategic missile systems.

Today, DRDL continues to drive innovation by contributing to advanced and next-generation missile systems such as hypersonic weapons, interceptor missiles, and long-range strategic deterrents. In alignment with DRDO's vision of self-reliance in defence technologies, DRDL actively collaborates with other DRDO laboratories, academic institutions, and Indian industries. Its continued commitment to technological excellence plays a vital role in enhancing India's defence preparedness and securing its strategic autonomy.

Introduction

In the current era of data-driven corporate management, the ability to transform raw data into actionable insights is a vital asset for any organization. This project, titled "Secure Employee Insights Dashboard," was developed during my internship to address the need for real-time monitoring of workforce data. By bridging the gap between a robust Oracle Database and the visualization capabilities of Grafana, the system allows stakeholders to monitor employee demographics, departmental distributions, and organizational growth through an intuitive, automated interface.

The core of this project lies in its modern backend architecture. Developed using the Spring Boot framework and managed via Maven, the application serves as a secure middleware that communicates directly with the Oracle database. Unlike traditional reporting methods that rely on manual exports, this solution utilizes a RESTful API to fetch data dynamically. This ensures that the information displayed on the dashboard is always current, providing a "single source of truth" for human resource metrics and organizational health.

The training report is based on a 6-month internship at DRDL, Kanchanbagh, Hyderabad. During this internship, a significant emphasis was placed on enterprise-grade security and professional development standards. To protect sensitive employee information, the connection between the backend and the dashboard is secured using a Service Account and Bearer Token authorization. This ensures that only authorized requests from Grafana are processed by the Spring Boot server. By implementing this machine-to-machine authentication protocol, the project not only delivers high-quality data visualization but also adheres to the strict cybersecurity standards essential for a premier organization like DRDL.

Objective of the Internship

The primary objective of this internship was to design and implement a secure, end-to-end data visualization pipeline that converts static database records into dynamic business intelligence. The project was designed to bridge the gap between backend data management and frontend observability, focusing on the following core goals:

- **Full-Stack Integration:** To gain hands-on experience in connecting a **Spring Boot** application to an **Oracle Database**. The goal was to master the use of JDBC (Java Database Connectivity) to execute optimized SQL queries and retrieve real-time employee data.
- **Implementation of Enterprise Security:** To understand and apply professional security standards by setting up **Service Account** authentication. A key objective was learning how to secure REST endpoints using **Bearer Tokens**, ensuring that sensitive HR data is only accessible to authorized systems and protected from unauthorized access.
- **Data Transformation and API Development:** To develop proficiency in building RESTful APIs that transform relational database rows into structured JSON format. This involved learning how to design efficient endpoints that **Grafana** can consume seamlessly for real-time reporting.
- **Professional Observability and UI Design:** To master **Grafana** as a visualization tool. The objective was to create a user-friendly dashboard that translates complex datasets into "glanceable" metrics, such as headcount gauges and departmental bar charts, while ensuring the interface remains responsive across various devices and screen sizes.

To gain valuable exposure to a professional working environment at **DRDL**, understanding the rigorous standards, security protocols, and collaborative workflows required within a premier defence research organization.

Overview of Project Name

The **Secure Employee Data Visualization System** is a specialized software solution designed to provide real-time visibility into personnel and departmental data. Developed during my tenure at DRDL, the project serves as a bridge between high-security data storage and modern analytical tools. Its primary function is to eliminate the need for manual reporting by automating the extraction, transformation, and visualization of employee records.

The system operates by leveraging a Spring Boot backend to interact with a centralized Oracle Database. Instead of raw database access, which can be a security risk, this project exposes specific, filtered data through a controlled API. This architecture ensures that sensitive information is processed and aggregated on the server side before being sent to the visualization layer, maintaining data integrity and reducing the load on the primary database.

The final component of the project is the Grafana Dashboard, which acts as the frontend interface. By utilizing a Service Account for secure communication, the dashboard pulls live metrics from the backend. This allows for the creation of interactive visualizations—such as hiring trends, department-wise distribution, and employee status gauges—providing a comprehensive, real-time overview of the organization’s workforce in a secure and professional format.

Tools and Technologies Used

The development of the Secure Employee Insights Dashboard utilized a modern full-stack architecture. The following technologies were integrated to build a secure, responsive, and high-performance pipeline:

1. Frontend (User Interface)

- React.js: A powerful JavaScript library used for building a dynamic and component-based user interface. It handled the state management and rendering of the dashboard components.
- Tailwind CSS: A utility-first CSS framework used for styling. It allowed for rapid UI development and ensured a modern, clean, and highly responsive design across all screen sizes.
- Axios / Fetch API: Used within React to make authorized asynchronous requests to the Spring Boot backend.

2. Backend Framework & Build Tools

- Java (JDK 17+): The primary programming language used for developing the robust backend logic.
- Spring Boot: The core framework used to build the standalone REST API. It provided the necessary modules to handle data requests and security filters.
- Maven: Used as the project management and build automation tool to handle the project lifecycle and dependencies via the pom.xml.

3. Database & Visualization

- Oracle Database: The enterprise-grade relational database used to store all employee-related records securely.
- Grafana: The visualization platform used to design interactive dashboards. In this project, Grafana panels were likely embedded or linked within the React frontend for a unified user experience.
- Oracle JDBC Driver: Facilitated the communication between the Spring Boot application and the Oracle server.

4. Security & Authentication

- Service Account: A dedicated account created to represent the application client.
- Bearer Token (Authorization): The security mechanism used for machine-to-machine authentication. The backend was configured to validate this token in the HTTP headers (e.g., Authorization: Bearer <token>).

System Design and Project Overview

The system architecture is designed as a secure, high-performance data pipeline. It follows a Decoupled Three-Tier Architecture, which separates the user interface, the business logic, and the data storage. This design ensures that each component can be maintained and scaled independently while maintaining high security standards.

1. Architectural Components

- **Frontend Tier (Presentation Layer)**: Developed using React.js and styled with Tailwind CSS, this layer is responsible for the user experience. It provides a responsive dashboard where users can view employee metrics. The frontend handles the client-side logic, such as sending requests to the backend with the necessary Authorization Tokens and rendering the data into visual components.
- **Backend Tier (Application Layer)**: The core engine of the project is built with Spring Boot. This tier acts as a "Secure Resource Server." It contains the API endpoints that the React frontend calls. It is responsible for:
    1. Validating the Service Account Bearer Token.
    2. Connecting to the Oracle Database via JDBC.
    3. Executing business logic and transforming SQL result sets into JSON format.
- **Database Tier (Data Layer):** The Oracle Database serves as the central repository. It stores structured tables containing employee information, departmental records, and historical data. Access to this tier is strictly limited to the Spring Boot application, ensuring that the database remains hidden from the public internet.

2. Data Flow and Integration

The workflow of the system follows a secure request-response cycle:

1. Authentication: When a user accesses the dashboard, the React application attaches a Service Account Token to the HTTP request header.

2. API Request: An asynchronous call (using Axios or Fetch) is made to the Spring Boot REST endpoint.
3. Security Validation: The Spring Boot backend intercepts the request, verifies the Bearer Token, and checks if the Service Account has the required permissions.
4. Database Query: Upon successful validation, the backend executes an optimized SQL query on the Oracle DB.
5. JSON Response: The backend packages the results into a lightweight JSON object and sends it back to the frontend.
6. Visualization: React receives the data, and the dashboard (utilizing Tailwind CSS for layout and potentially embedded Grafana frames for complex charts) updates instantly to show the live employee insights.